Movie App: How AI Assisted, and Manual Improvements

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Deliverable 3: How AI assisted

I used Antigravity CLI with Gemini 3.8 Flash as my development assistant. I planned first (Prompt 1 asked for a plan and no code), then built one step per prompt: setup, OMDB client, home screen, router and header, search, favorites, card polish, and Firebase login with per-user favorites. All prompts and logs are in PROMPTS.md. Prompt numbers differ from step numbers: Prompt 11 is Step 8a, Prompt 14 is Step 9b-i, Prompt 15 is Step 9b-ii and Prompt 16 is Step 10.

From Prompt 5 onward, every prompt named the files the AI could touch, required it to propose each edit and wait for my approval, and forbade commands I had not asked for. I added these rules after Step 3, when the AI edited three files after I approved and ran a type check I had not requested. I reviewed every code change before committing, and I committed AI output separately from my own edits.

What worked. The MVVM split kept the code easy to explain: models in pure TypeScript, view-model hooks for state, and JSX-only views. The models were written defensively, so invalid JSON, non-array values and bad entries are filtered out (FavoritesModel and UserDataModel). CSS changes were append-only from Step 4 onward, which I confirmed from the diff each time (84, 49, 46, 145 and 14 insertions, 0 deletions). For the Firebase favorites I specified the sync rules and the AI implemented them: save only after a successful load (Prompt 15) and no rollback on a failed save (Prompt 16). The trade-off is that a failed save shows an error message but the list is not rolled back, and I could not test that message safely on a live project.

Where it fell short. It missed the "N/A" poster value and dead poster URLs, so I fixed those by hand (commits 52738fa and 8239b33). It labelled append-only CSS as "Create" in Steps 6b, 8b-ii and 10; each time I checked with git diff --stat or --numstat and found a pure append. In Step 7a it repeated a CSS rule that already existed, despite my instruction to re-read the file, and in Step 7b it wrote a ref during render. Its summaries were claims that I had to verify from disk.

Checking the AI. On Prompt 11 the output included a Sources list from unrelated repositories, and it read files I had not named. I added "Do not search the web" to Prompt 12, and no Sources list appeared on Prompts 12 and 13. On Prompt 14 it came back, with seven GitHub entries, while the AI's summary said no web searches were run. I had also approved that prompt with a single word, so from Prompt 15 I used full approval wording and put the rule on the first line as well as at the end. No Sources list appeared on Prompts 15 and 16. That is two prompts, so it is encouraging, not proof.

My part. I chose the architecture, wrote the prompts, reviewed the diffs and ran the tests. Every step that changed behaviour was tested in the browser, including corrupt localStorage, keyboard use, an offline reload and two accounts, to check that one user cannot see another's favorites. I tested the Firestore rules in the simulator (four simulations) before publishing them. Prompts 11, 12 and 14 added no visible behaviour, so for those I checked that the app still loaded, ran scans and ran the build. The API key stayed out of the AI session and in a git-ignored .env file, and I checked results against the live OMDB API.

Deliverable 4: Manual improvements, corrections and refactoring

My hand edits were in Steps 2 and 3. Steps 4 to 10 needed no hand edits to the code, so the table also lists a refactor I directed after review, and the process corrections follow it.

#	File	Problem found	What I did	Commit
1	HomeModel.ts	toMovie passed OMDB's "N/A" poster string through as a URL, so cards showed a broken image.	Set poster to an empty string when OMDB returns N/A, so the card shows its "No poster" placeholder.	52738fa
2	HomeModel.ts	The shuffle comment said "in place" but the code returns a copy.	Corrected the comment.	52738fa
3	main.tsx	Temporary Step 2 test code was still in the file.	Restored the file to its Step 1 version.	aac8622

#	File	Problem found	What I did	Commit
4	MovieCard.tsx	Some poster URLs are valid but dead, so the card showed a broken image icon. I found this by looking at the live grid.	Added imageFailed state and an onError handler that falls back to the placeholder.	8239b33
5	FavoritesContext.tsx	Review finding: toggle saved inside the setFavorites updater, which React expects to be pure.	Directed a refactor (Prompt 10) so toggle computes the list from a ref, saves once, then sets state. The AI implemented it.	4138b06

Process corrections

- My first edits to HomeModel.ts used line numbers in Notepad and merged lines. git diff showed the damage before I committed, git restore recovered the file, and I redid the edits with find-and-replace.
- A stray instruction was committed into PROMPTS.md because I had checked only the file count (35ac416). I read the diff, rebuilt the file and fixed it (b4d493e).
- Prompt 7 contradicted itself about where FavoritesModel.ts should live. The AI asked instead of guessing, and I chose src/models/.
- I approved Prompt 14 with one word, which does not restrict further edits or commands. From Prompt 15 I used the full approval wording.

Left in on purpose and logged as candidate improvements: the redundant CSS rule from Step 7a, the ref written during render in Step 7b, the effect that depends on the user object instead of user.uid, both handleSubmit and an effect navigating after login, and the 809 kB bundle with no code-splitting.